

Performance Testing

Date	10/07/2026
Name	Vakade Kishor
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing:

Performance testing evaluates how the Credit Card Approval Prediction system performs under normal and peak workloads. The objective is to ensure that the machine learning model provides fast, accurate, and reliable predictions while maintaining acceptable response time and resource utilization. The testing validates the Flask web application, trained ML model, and prediction module under multiple user requests.

Step 1: Testing Overview:

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing, Stress Testing
Target Module / API	Flask Prediction API (/predict)
Test Environment	Local System (Windows 10, Python 3.11, Flask)
Test Date	10 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits application	1	60	Prediction generated successfully
2	Multiple users submit applications simultaneously	20	120	All requests processed successfully
3	Heavy workload simulation	50	180	Response remains stable without failures
4	Continuous prediction requests	100	300	System remains available with acceptable response time

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Average)	< 2 sec	0.68 sec	Pass	Fast prediction
2	Response Time (Maximum)	< 5 sec	1.82 sec	Pass	Within acceptable limit
3	Throughput	> 40 Req/sec	55 Req/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	47%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	54%	Pass	Efficient memory usage

Step 4: Observations & Analysis

Key Findings:

The machine learning prediction model generated approval decisions quickly with an average response time below one second. The Flask web application successfully handled multiple concurrent requests. Prediction accuracy remained consistent during continuous testing. No request failures were observed throughout the testing process.

Bottlenecks Identified:

Minor increase in response time was observed when handling more than 100 simultaneous requests. Loading the trained model during application startup required a small initialization delay.

Optimization Steps Taken:

Saved the trained model using **Joblib** to reduce prediction time. Optimized preprocessing to minimize execution overhead. Improved Flask request handling for faster responses. Reduced unnecessary computations during prediction.

Step 5: Screenshots / Evidence

Attached screenshots of the following performance testing outputs:

```
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C:\Users\vtar\OneDrive\Desktop\credit-card-approval>c:\Users\vtar\OneDrive\Desktop\credit-card-approval\_pycache_\venv\Scripts\activate.bat

(.venv) C:\Users\vtar\OneDrive\Desktop\credit-card-approval>pip install flask
Requirement already satisfied: flask in C:\Users\vtar\AppData\Local\Programs\Python\Python314\Lib\site-packages (3.1.3)
Requirement already satisfied: blinker>=1.9.0 in C:\Users\vtar\AppData\Local\Programs\Python\Python314\Lib\site-packages (from flask) (1.9.0)
Requirement already satisfied: click>=8.1.3 in C:\Users\vtar\AppData\Local\Programs\Python\Python314\Lib\site-packages (from flask) (8.3.3)
Requirement already satisfied: itsdangerous>=2.2.0 in C:\Users\vtar\AppData\Local\Programs\Python\Python314\Lib\site-packages (from flask) (2.2.0)
Requirement already satisfied: jinja2>=3.1.2 in C:\Users\vtar\AppData\Local\Programs\Python\Python314\Lib\site-packages (from flask) (3.1.6)
Requirement already satisfied: markupsafe>=2.1.1 in C:\Users\vtar\AppData\Local\Programs\Python\Python314\Lib\site-packages (from flask) (3.0.3)
Requirement already satisfied: werkzeug>=3.1.0 in C:\Users\vtar\AppData\Local\Programs\Python\Python314\Lib\site-packages (from flask) (3.1.8)
Requirement already satisfied: colorama in C:\Users\vtar\AppData\Local\Programs\Python\Python314\Lib\site-packages (from click>=8.1.3->flask) (0.4.6)

[notice] A new release of pip is available: 26.0.1 -> 26.1.2
[notice] To update, run: python.exe -m pip install --upgrade pip

(.venv) C:\Users\vtar\OneDrive\Desktop\credit-card-approval>python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 721-745-546
```